(a) Centrifugation (b) Dialysis
(c) Reverse osmosis (d) Diffusion

Ans. (a) : The mechanical process which involves the use of centrifugal force to separate particles from a solution according to their seize, shape density, medium viscosity and rotor speed is known as centrifugation. The working

CHEMISTRY

97. The characteristic odour of garlic is due to :
(a) A chloro compound
(b) A sulphur compound
(c) A florine compound
(d) Acetic acid

Ans. (b) : The reason for the characteristic smell of garlic is due to the 0.1% essential oil present in it. The main constituents of this oil are diallyl disulphide, diallyl trisulphide and ally propel disulfide, which are all compounds of sulphur.

98. Which one of the following is an active component of oil clove?
(a) Menthol (b) Eugenol
(c) Methanol (d) Benzaldehyde

Ans. (b) : Oil of cloves, also known as Clove oil, is an essential oil obtained from the clove plant. It is a natural analgesic and antiseptic used primarily in dentistry for its main ingredient eugenol. The oil produced by cloves can be used in many things from flavouring medicine to remedies for bronchitis, common cold, a cough, fever, sore throat and tending to infections.

99. The most reactive among the halogens is :
(a) Fluorine (b) Chlorine
(c) Bromine (d) Iodine

Ans. (a) : The most reactive gas among halogens is fluorine. Fluorine (F) has the most negative electric charge of all the other halogens. Due to this quality, it is most active. The order of their reactivity is as follows - Fluorine → Chlorine → Bromine → Iodine .

100. Which one of the following was used as a chemical weapon in the First World War?
(a) Carbon monoxide (b) Hydrogen cyanide
(c) Mustard gas (d) Water gas

Ans. (c) : Mustard gas is a poisonous gas which is powerful as well as explosive. Its chemical name is dichloro-diethyl -sulfide [S(CH,CH,CI] .It is a colorless gas. Mustard gas was used as a chemical weapon in World War I, while gases called Phosphine and Sarin were used in World War II.

101. Which one of the following is paramagnetic in nature?
(a) Iron (b) Hydrogen
(c) Oxygen (d) Nitrogen

Ans. (c) : Oxygen is paramagnetic because the oxygen molecule contains two unpaired electrons. Hence, option C is correct.

102. Which one of the following has the highest fuel value?
(a) Hydrogen (b) Charco
(c) Natural gas (d) Gasoline

Ans. (a) : Fuel value can be expressed in terms of calorific value of fuel. The calorific value of a fuel is the amount of heat produced by burning 1 kg of fuel. Hydrogen has the highest calorific value (141,790 KJ/kg) thus have highest fuel value. Calorific value of charcoal, natural gas and gasoline are (29,600) 43,000 47,300 kJ/kg) respectively. Natural gas majorly consists of methane.

103. Which one of the following is used as an anti-freeze for the automobile engines?
(a) Propyl alcohol (b) Ethanol
(c) Methanol (d) Ethylene glycol

Ans. (d) : Ethylene glycol is used as an antifreeze in automobile engines. It is prepared by the oxidation of ethylene. Where the cold is more, alcohol is used in place of mercury for the thermometer. Whereas ethylene glycol is used in the form of antifreeze in vehicles.

104. The major component of honey is:
(a) Glucose (b) Sucrose
(c) Maltose (d) Frustose

Ans. (d) : The major component of honey is fructose. Composition of honey (in percentage) is.

Fructose	–	38.2
Sucrose	–	1.5
Glucose	–	31
Minerals	–	0.5
Water	–	17.1
Maltose	–	7.2
Carbohydrate	–	4.2

BIOLOGY

105. The pituitary gland by virtue of its tropic hormones controls the secretary activity of other endocrine glands. Which one of the following endocrine glands can function independent of the pituitary gland.
(a) Thyroid (b) Gonads
(c) Adrenals (d) Parathyroid

Ans. (d) : Parathyroid gland is such a gland, which works completely independently. The hormones secreted by the pituitary gland do not play any role in its activity. The pituitary gland is also called the master gland because it controls the secretion of the endocrine

glands. The personality of a person also motivates health, growth and sexual development.

106. Oxygen transportation in a human body takes place through?
1. Blood 2. Lungs 3. tissue
The correct sequence of transportation is
(a) 1, 2, 3 (b) 3, 1, 2
(c) 2, 1, 3 (d) 1, 3, 2

Ans. (c) : The first organ in the human body where oxygen is acquired is the lungs. Oxygen enters the blood after the lungs and uptake from the blood to the tissues. Oxygen is mainly transported by the red pigment found in blood, hemoglobin. Hemoglobin is purple in color whereas oxy hemoglobin is bright red in colour. The amount of oxygen that will combine with hemoglobin is based on the pressure of oxygen and the pH of the blood.

107. Corpus luteum is a mass of cells found in:
(a) brain (b) ovary
(c) pancreas (d) spleen

Ans. (b) : Corpus luteum is a mass of cells found in ovary. It produces progesterone hormone. If the ovary is not fertilised, the corpus-luteum degenerates after about two weeks and is gradually replaced by ordinary ovary tissue.

108. Alpha-keratin is a protein preset in:
(a) blood (b) skin
(c) wool (d) eggs

Ans. (b) : Alpha-keratin, or α-keratin, is a type of keratin found in vertebrates. This protein is the primary component in hairs, horns, mammalian claws, nails and the epidermis layer of the skin α-keratin is a fibrous structural protein, meaning, it is made up of amino acids that form a repeating secondary structure.

109. What is the average fat content of buffalo milk?
(a) 7.2% (b) 4.5%
(c) 9.0% (d) 1.0%

Ans. (a) : Average fat content in buffalo milk is 7.2% and average fat content in cow milk is 4.4%. As per the nutrient components, buffalo milk contains all the nutrients in higher proportion than cow's milk.

110. Which one of the following techniques can be used to establish the paternity of a child?
(a) Protein analysis
(b) Chromosome counting
(c) Quantitative analysis of D.N.A.
(d) D.N.A. finger printing

Ans. (d) : DNA finger printing technology is the process of establishing the biological paternal relationship between individual and his/her child on the analysis of sample cells taken from each of them. DNA finger printing can solve cases of rape murder etc.

111. Arteries supplying blood to the heart are called:
(a) Carotid arteries
(b) Hepatic arteries
(c) Coronary arteries
(d) Pulmonary arteries

Ans. (c) : The arteries which supply blood to the heart are known as coronary arteries. The right coronary artery supplies blood to right side of heart while left coronary artery supplies blood to left side of heart. Hepatic artery is a short blood vessel that supplies oxygenated blood to the liver, pylorus (a part of the stomach), duodenum (a part of the small intestine) and pancreas. The carotid arteries are major blood vessels in the neck that supply blood to the brain, neck, and face. Thus, correct answer is option C.

112. Recommended daily intake of proteins for a moderately active women in
(a) 30 g (b) 37 g
(c) 40 g (d) 46 g

Ans. (d) : Daily intake of protein for active woman is 46 gram. But a pregnant women's protein requirement increase to 47 – 60 gm per day.

113. Match List-I with List-II and select the correct answer:

List-I	List-II
A. Malaria	1. Bone-marrow
B. Filaria	2. Brain
C. Encephalitis	3. Muscle
D. Leukaemia	4. Lymph node
	5. Blood cells

Codes:

	A	B	C	D
(a)	5	3	2	1
(b)	5	4	2	1
(c)	4	3	5	1
(d)	5	4	1	2

Ans. (b) : The correct match is mentioned as below–

Malaria	–	Blood cells
Filaria	–	Lymph node
Encephalitis	–	Brain
Lukaemia	–	Bone marrow

ENVIRONMENT AND ECOLOGY

114. Which one of the following organisms is likely to show the highest concentration of D.D.T.